FINAL YEAR PROJECT REPORT
UCI HAR & UniMiB SHAR

COMPARISON BETWEEN THE TWO DATASETS

Aspect	UCI HAR	UniMiB SHAR
Purpose	General human activity recognition	Fall detection + daily activity recognition (health-oriented)
Devices Used	Samsung Galaxy S II smartphone	Samsung Galaxy Nexus smartphone
Sensor Placement	Fixed: phone worn on the waist	Variable but typically in front trouser pockets
Sensors	Accelerometer + Gyroscope (3-axis each)	Accelerometer only (3-axis)
Sampling Rate	**50 Hz**	**50 Hz**
Activities	6 Activities: walking, walking up/down, sitting, standing, laying	17 daily activities + 8 fall types (total 25 classes)
Data Format	Pre-segmented windows (2.56 s, 128 samples)	Raw accelerometer windows around events (≈3s)
Participants	30 subjects	30 subjects
Labels	Clean, structured activity labels	Activity + fall category labels
Participants	30 (Males)	30 (6 male and 24 female)

ISSUES WITH THESE DATASETS:

- Even dropping a phone will be captured as a type of Fall.
- **Distinguishing *Types* of Falls :** The difficulty in distinguishing between the **different types of falls,** because the acceleration is more or less the same.
- All the stationary activities such as SITTING, STANDING AND LAYING makes it difficult to predict them properly.

RESEARCH PAPERS :

S.NO	PAPER
1	Sekaran, Sarmela Raja; Pang, Ying Han; You, Lim Zheng; Yin, Ooi Shih (2024). Description of UCI HAR and UniMiB SHAR.. PLOS ONE. Dataset. https://doi.org/10.1371/journal.pone.0304655.t001 - Proposes a model that integrates TCN-GRU
2	Debashish Saha;Piyush Malik;Adrika Saha (2025).Activity and Subject Detection for UCI HAR Dataset with & without missing Sensor Data. rXiv:2505.06730v1 [cs.LG] 10 May 2025 -
3	Farhad Mortezapour Shiri, Thinagaran Perumal, Norwati Mustapha and Raihani Mohamed (2025); Deep Learning and Federated Learning in Human Activity Recognition with SensorData:A Comprehensive Review

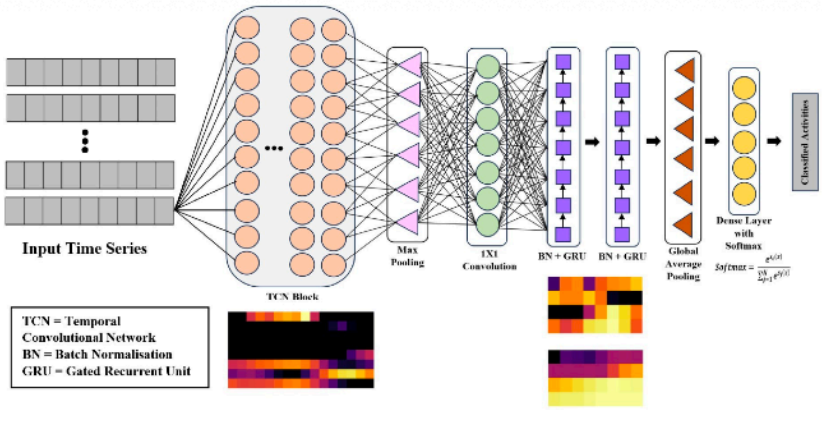
PAPER 1 :

About:

The paper focuses on recognising human activities (walking, sitting, standing, falls, etc.) from smartphone accelerometer and gyroscope signals without manual feature engineering. The goal is to design a model that keeps high accuracy while using relatively few parameters and low computation, making it suitable for resource-constrained devices.

Proposed Solution:

Designed and evaluated a hybrid network that combines Temporal Convolutional Network (TCN) with Gated Recurrent Unit (GRU) layers to extract spatiotemporal features from raw inertial time series. This technique reduces the cost drastically when compared with the cost associated with LSTM.



RESULT:

- The proposed TCN-GRU achieves very high accuracy with few parameters: about **97.25%** on UCI HAR and **93.51%** on UniMiB-SHAR using only around 0.18–0.19 million trainable parameters, outperforming or matching many heavier CNN, LSTM, and TCN baselines.

TABLE I
ACCURACY WITH SIMULATED MISSING DATA AND IMPUTATION FOR ACTIVITY RECOGNITION

Missing data intervals	Accuracy with missing data	SI	KNN	SI+PCA	KNN+PCA
1s ACC 1s Gyro (5 intervals)	0.7822	0.8989	0.8860	0.8975	0.8856
5s ACC 5s Gyro (1 interval)	0.7872	0.8901	0.8931	0.8907	0.8928
5 Sec ACC (2 intervals)	0.7716	0.8663	0.9233	0.8653	0.9237
5s Gyro (2 intervals)	0.7974	0.9189	0.9301	0.9186	0.9284
10s ACC (1 intervals)	0.7710	0.8626	0.9138	0.8612	0.9138
10s Gyro (1 interval)	0.7740	0.9308	0.9362	0.9298	0.9348

TABLE II
ACCURACY WITH SIMULATED MISSING DATA AND IMPUTATION FOR SUBJECT RECOGNITION

Missing data intervals	Accuracy with missing data	SI	KNN	SI+PCA	KNN+PCA
1s ACC & 1s Gyro (5 intervals)	0.6150	0.6417	0.7340	0.6282	0.7199
5s ACC & 5s Gyro (1 interval)	0.6184	0.6383	0.7340	0.6262	0.7170
5s ACC (2 intervals)	0.6180	0.6413	0.7442	0.6277	0.7282
5s Gyro (2 intervals)	0.6141	0.6510	0.7515	0.6403	0.7398
10s ACC (1 interval)	0.6155	0.6340	0.7476	0.6233	0.7340
10s Gyro (1 interval)	0.6170	0.6490	0.7587	0.6379	0.7481

PAPER 2 :

About:

- Focuses on recognizing the individuals performing activities, which is important for personalized and context-sensitive applications.
- It explores the issue of missing sensor data, a common real-world problem due to hardware malfunctions or communication faults, which significantly affects the accuracy and reliability of HAR models.

PROPOSED SOLUTION:

LSTM network is used and trained on UCI HAR data (10,299 samples, 561 features). Missing data is imputed via SimpleImputer (mean/median) or KNNImputer.

Suggested sensors – compasses, gyroscopes, magnetometers, and accelerometers

RESULT:

activity: 93.89% accuracy, subject: **80.19%** plus KNN imputation for missing data recovery (activity up to **93.62%**, subject up to **75.87%** without PCA).

General Issues :

- **CHA – Complex Activity Recognition :**
  Few papers have proposed few techniques to capture them but in-efficient, techniques such as Latent Semantic Analysis

- **Multi-resident Activity Recognition :**
  Wearable-Based Approaches: These methods require

residents to wear personal sensors, like smart jewelry or wristbands, which can provide precise data to associate activities with specific individuals.

Data Driven Approaches: These techniques treat data association as a distinct learning task that precedes activity classification, allowing for more a nuanced understanding and categorisation of activities.

Single ModelApproaches: Leveraging raw ambient sensor data, these models implicitly learn data associations during the training process, aiming to integrate and correlate data without explicit labelling.

Applications :

- AAL (Ambient Assisted Living) – More Helpful in remote monitoring of the elderly people with dementia and other related health conditions.

- Healthcare Systems – e-health systems applications, such as remote patient care, respiratory biofeedback, comprehensive activity monitoring, mental stress evaluation, both physical and mental rehabilitation, weight training, as well as real-time assessment of posture and movement.

- Fitness and Sports :
  Traditionally,dietitians and medical professionals have relied on self-completed questionnaire methods, asking participants to log their daily activities. These questionnaires are then analyzed to assess the individual's physical activity level and provide tailored feedback. However,the analysis ofself-reported data is time-consuming and labor-intensive, especially with large populations. To overcome these challenges, recent advances have seen the adoption of sensor-based technologies that provide a more effective means of capturing and compiling daily contextual data and lifelogs. Moreover, in the area of sports, the objective

evaluation of an athlete's performance is crucial. Automated sport-specific movement detection and recognition systems, facilitated by sensor-based HAR technologies, provide significant improvements over manual performance analysis methods. These systems are employed during training sessions and competitions across various sports.

- Security and Surveillance

Table 1 (continued)

Dataset name	Year	Environment	Type and number of sensors	Number of activities	Duration	Multi-resident
CSL-SHARE [142]	2021	Lab	Wearable sensors (Accelerometers, Gyroscopes, EMG, Electrogoniometer, Airborne microphone) (10)	22	2 h	No
SaMO-UJA [143]	2021	Smart lab	Ambient sensors + BLE beacons (85)	25	9 days	Yes
HARTH [144]	2021	Lab	Wearable (Accelerometer) (2)	10	2 h	No
KU-HAR [145]	2021	Lab	Accelerometers and Gyroscopes embedded in smartphone	18	–	No
SDHAR-HOME [105]	2022	1 Smart home	Wearable + Ambient sensors (35) + Positioning (7)	18	2 Months	Yes
HAR70+ [146]	2023	Lab	AX3 accelerometers	6	–	NO

Table 1 (continued)

Dataset name	Year	Environment	Type and number of sensors	Number of activities	Duration	Multi-resident
CASAS: Milan [110]	2009	Smart home	Ambient sensors (motion, temperature, door closure) (33)	15	72 days	Yes
CASAS: Cairo [110]	2009	Smart home	Ambient sensors	13	–	Yes
OPPORTUNITY [111]	2010	Lab	Wearable + Ambient sensors (72)	18	25 h	No
CASAS: Kyoto Multi-resident [112]	2010	Smart home	Ambient sensors (motion, item, cabinet, water, temperature) (37)	15	2 months	Yes
HASC [113]	2011	Lab	Inertial sensors embedded in smartphone	6	–	No
VanKasteren [114]	2011	3 Smart home	Ambient and wireless sensors	10-13-16	25-14-19 days	Yes
PAMAP2 [115]	2012	Lab	Inertial (accelerometers, Gyroscope, Magnetometer), heart rate sensor	18	10 h	No
CASAS: Tulum [116]	2012	2 Smart home	Ambient sensors	16	–	Yes
CASAS: Aruba [116]	2012	Smart home	Ambient sensors (34)	11	220 days	No
USC-HAD [117]	2012	Lab	Accelerometer, Gyroscope, Magnetometer	12	6 h	No
RealDISP [118]	2012	Lab	Wearable sensors (Accelerometers, Gyroscopes, Magnetometer) (9)	33	9 h	No
UCI-HAR [119]	2013	Lab	Accelerometers and Gyroscopes embedded in smartphone	6	–	No
ARAS [120]	2013	2 Smart home	Ambient sensors (20)	27	2 months	Yes
WHARF [121]	2013	Lab	Accelerometer	12	–	No
DSADS [122]	2014	Lab	Magnetometers, Inertial sensors (5)	19	–	no
MHEALTH [123]	2015	Controlled	Wearable sensors (Accelerometers, Gyroscopes, MAGNETOMETER, ECG)	12	–	No
HHAR [124]	2015	Controlled	Accelerometers and Gyroscopes embedded in smartphone and smartwatch	6	–	No
UTD-MHAD1 [125]	2015	Lab	Accelerometer, Gyroscope	27	–	No

Table 1 (continued)

Dataset name	Year	Environment	Type and number of sensors	Number of activities	Duration	Multi-resident
UTwente [126]	2016	Lab	Accelerometer, Gyroscope, and Linear Acceleration embedded in two mobile phones	13	–	No
REALWORLD [127]	2016	Controlled	Wearable sensors (Accelerometers, Gyroscopes, Magnetometer)	8	–	No
AreM [128]	2016	Lab	Inertial sensors embedded in the smartphone, wearable sensors	7	–	No
MobiAct 1 [129]	2016	Lab	Accelerometer, Gyroscope, Orientation Sensors embedded in smartphone	9	–	No
MobiAct 2 [130]	2017	Lab	Inertial sensors (Accelerometers, Gyroscope)	12	–	No
UniMiB SHAR [131]	2017	Lab	Accelerometers embedded in smartphone	9 + 8	–	No
ExtraSensory [132]	2017	Lab	Accelerometers, Gyroscope, GPS embedded in smartphone	10	–	No
ContextAct@A4H [133]	2017	Smart home	Ambient sensors + Actuators (219)	7	1 month	No
SIMADL [134]	2018	Smart home	Ambient sensors (29)	5	63 days	No
DU-MD [135]	2018	Lab	Wearable (accelerometer)	7	–	No
HuGaDB [136]	2018	Lab	Inertial (Accelerometers, Gyroscope)	12	10 h	No
MotionSense [137]	2018	Lab	Accelerometers and Gyroscopes embedded in smartphone	6	–	No
WISDM [138]	2019	Controlled	Accelerometers and Gyroscopes embedded in smartphone and smartwatch	18	60 h	No
Ulster [139]	2019	Lab	Accelerometer, Gyroscope, GPS, light sensors, embedded in smartphone and smartwatch	6	6 weeks	No
W-HAR [140]	2020	Controlled	Inertial (Accelerometers, Gyroscope) and Stretch sensors	7	–	No
MARBLE [141]	2021	Lab	Wearable + Ambient Sensors (8)	13	16 h	No